Assignment 2: Apply a published method to a new dataset

Individual assignment

The assignment

You must produce a report that answers the biological question, by explicitly addressing the question below. Your report must link to your code which must be made accessible to the teacher.

The question

Golub et al. (1999) classified leukemia samples from gene-expression measurements using a small set of weighted features. You will apply the published weighted-voting method to a different disease and measurement system: numerical properties extracted from breast fine-needle-aspirate images.

> Can the Golub weighted-voting method classify malignant breast samples accurately and robustly?

This is an applicability study, not only a coding exercise. Your answer must connect what was measured, what the algorithm receives, how patients are held out, what the metrics show, and what biological or clinical claim remains unsupported.

Supplied material

- `data/wdbc.csv`: frozen Wisconsin Diagnostic Breast Cancer data;
- `data/MANIFEST.json`: provenance, class counts, and SHA256;
- `DATA_DICTIONARY.md`: variables and measurement origin;

The input contains 569 patients and 30 numerical features. WDBC was developed in work by William H. Wolberg, W. Nick Street, and Olvi L. Mangasarian at the University of Wisconsin and was donated to the UCI repository in 1995. It addressed a diagnostic question: **can quantitative nuclear morphology extracted from a digitized fine-needle-aspirate slide distinguish a malignant from a benign breast mass?** The associated 1994 study describes these 569 patients as a consecutive development series. The cases had already undergone sampling of a breast mass; this was not a population-screening dataset.

One row is one patient/sample. The positive class is malignant (`1`); benign is `0`. The supplied table starts after image acquisition, cell-nucleus segmentation, and feature extraction: it contains neither patients' clinical histories nor the original images. The original article also reports 54 additional new patients used to test its system; those patients are not included

here. Read `DATA_DICTIONARY.md` for the measurement pipeline and source limitations. Do not replace the supplied file with another copy.

Your work

1. Data exploration

First, explore the data. Report its size, the different features, their link to the diagnosis label. This is a descriptive analysis that should not influence your feature selection.

2. Implement weighted voting

For feature j, compute the training-set weight

$$w_j = \frac{\mu_{j,1} - \mu_{j,0}}{\sigma_{j,1} + \sigma_{j,0}}.$$

Select the ten features with largest $|w_j|$. S is the set of selected features. For a sample x, compute

$$V(x) = \sum_{j \in S} w_j \left(x_j - \frac{\mu_{j,1} + \mu_{j,0}}{2} \right).$$

Predict malignant when $V(x) \geq 0$. Means, standard deviations, midpoints, weights, and selected features must be estimated from the training fold only.

3. Evaluate the model

Compute your personal seed as

$$\text{seed} = JJ + MM + YY,$$

where `JJ` is your day of birth, `MM` is your month of birth, and `YY` is the last two digits of your year of birth, all interpreted as integers. For example, `17/04/2003` gives $17 + 4 + 3 = 24$. Report only the resulting seed, not your birthdate.

You will now evaluate the capacity of the model (using weighted voting with ten training-selected features as implemented above) to predict the patient diagnosis, based on a cross-validation study.

Use five-fold stratified cross-validation repeated three times with your personal seed (hint: use *sklearn.model_selection.RepeatedStratifiedKFold*.

Take one model an exemple to present the confusion matrix, auc_score and balanced_accuracy (hint: use sklearn.metrics) then use average measures over the 15 models (3 repeats, 5 folds) to present the model performance.

4. Evaluate feature selection stability

Record the selected feature names in every outer fold. Report:

- selection frequency for every feature;
- mean pairwise Jaccard similarity of the 15 selected sets;
- whether stable prediction coincides with stable feature identity.

5. Compare

Compare the method with a prediction based on logistic regression.
Evaluate logistic regression on the same 15 train/validation splits and with the same personal seed as weighted voting. Use all 30 supplied features. Standardize features within each training fold (hint: use StandardScaler), never fit the scaler using validation data. Use sklearn LogisticRegression(solver="liblinear", C=1.0, max_iter=1000) with default class weighting and no hyperparameter tuning. Predict classes using the default probability threshold of 0.5.

6. Conclude on the clinical question

In the clinical context of using the method to predict the diagnosis of a patient, would you recommend this method? Elaborate on the limits and whether further analyses are needed.

7. Paper reproducibility

Comment on the reproducibility of the method from the paper based on this work. What should have been done to make it more reproducible.

Requirements

The report must at most 10 pages, excluding tables, figures and references in pdf format.

State your resulting personal seed, but do not state your birthdate.

Code implementation must be done in python using the uv framework for environment management

Include a `README.md` in your code that must give one command that runs your entire pipeline running all tables and figures.

Comment your code to explain implementation choices.

Cite Golub et al. (1999), the WDBC dataset, and any reused code.